

Vivek Dhavalkumar Patel

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EDUCATION

Virginia Polytechnic Institute and State University

Blacksburg, Virginia

Pamplin College of Business

Graduation: May 2028

- B.S in **Fintech and Big Data Analytics (Quantitative Finance)**
- **Related Coursework:** Intro to Accounting I & II, Calculus I – III, Business Statistics, Microeconomics, Macroeconomics

EXPERIENCE

Small Independent Business | Entrepreneurial Venture

Dec 2024 - Present

- Founded and scaled a personal business serving **50–70 recurring clients**, managing end-to-end operations including scheduling, customer relations, and service delivery.
- Leveraged social media and referral-based marketing to increase client acquisition by **~30%**, building a strong local brand presence.
- Generated approximately **\$10,000+** in revenue through consistent weekly appointments, pricing strategy, and a strong clientele.

Volvo Penta | Extern

Jun 2024 – Jul 2024

- Shadowed senior finance and operations professionals to analyze end-to-end manufacturing workflows, gaining exposure to cost structures, supply chain dynamics, and revenue drivers within the marine engine sector.
- Collaborated cross-functionally with project management and engineering teams to understand budget allocation, production timelines, and operational efficiency metrics.
- Assisted in evaluating business processes and identifying opportunities to improve workflow efficiency and cost optimization across departments.

PROJECTS

AI Zoning Agent | IBM Project | <https://zoning-agent-app-web.vercel.app/>

March 2026

- Built an **AI-powered conversational agent** using **IBM watsonx.ai** and **Google APIs** that lets users query complex municipal zoning regulations in natural language, cutting zoning research time by **50–60%** for non-technical users.
- Engineered a **Retrieval-Augmented Generation (RAG)** pipeline over local zoning ordinances to deliver accurate, context-aware answers for property compliance questions, grounding **LLM outputs** in authoritative source documents.
- Designed scalable data pipelines and optimized **NLP workflows** for **real-time retrieval**, improving response accuracy and consistency across diverse zoning scenarios and edge cases.

CONDUIT | AI Infrastructure & Commodity Intelligence Platform | conduit-vivek-patel.vercel.app/

May 2026

- Replaced a **linear** scoring formula with a **five-pillar composite engine** incorporating compound interaction penalties and geographic proximity weighting, trained on **34+** real hyperscale deployments across **Microsoft, Google, AWS, and Meta** using **React and Supabase**.
- Engineered **12 quantitative** features (convenience yield, futures curve slope, COT positioning) into Bidirectional **LSTM** and **XGBoost** models across **16 commodity markets**, improving directional prediction accuracy from **~62%** to **~76%**.
- Deployed full-stack application on Vercel and Railway with **sub-200ms** ML inference and Claude AI-powered commodity shock classification across **10+ live data** APIs (NASA POWER, EIA, USGS, NewsAPI, Alpha Vantage).

Arcanium Labs | Co-founder | Chesapeake, VA

August 2024 – May 2025

- Co-founded an AI startup to automate constructions estimating; **\$800K** raised and accepted into **Microsoft for Startups** with **\$350K in Azure credits**.
- Led technical teams to design and deploy custom AI agents for material takeoffs, logistics, and cost analysis.
- Returned all investment back to investors because I wanted to go to college.

LEADERSHIP

Phi Gamma Nu | Board Member

Feb 2026 – Present

- Spearheading the launch of a professional business fraternity chapter from the ground up, developing marketing strategies, recruiting members, and establishing fraternity presence on campus.

Technical Skills: Python (pandas, numpy, hmmlearn, yfinance), SQL, Java, Advanced Excel, Tableau, Financial Modeling, Time Series Analysis, Machine Learning, Statistical Modeling, Risk Analysis, Data Visualization, Power BI, React, Tensorflow

Certifications: Bloomberg Market Concepts, Claude Education, JPMorgan Investment Banking Forage Simulation

Interests: Tennis | Pickleball | Poker | Cooking | Photography | Music | Hiking